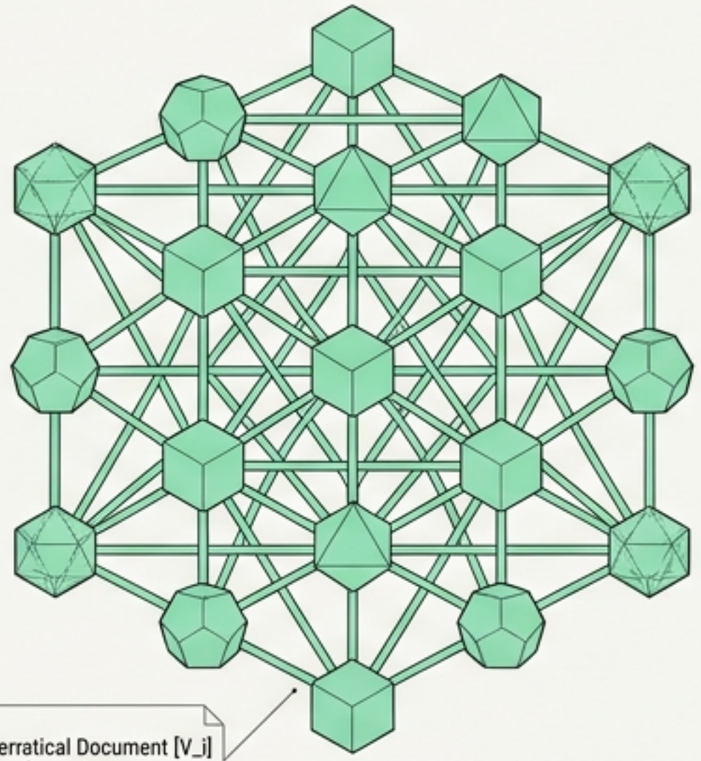
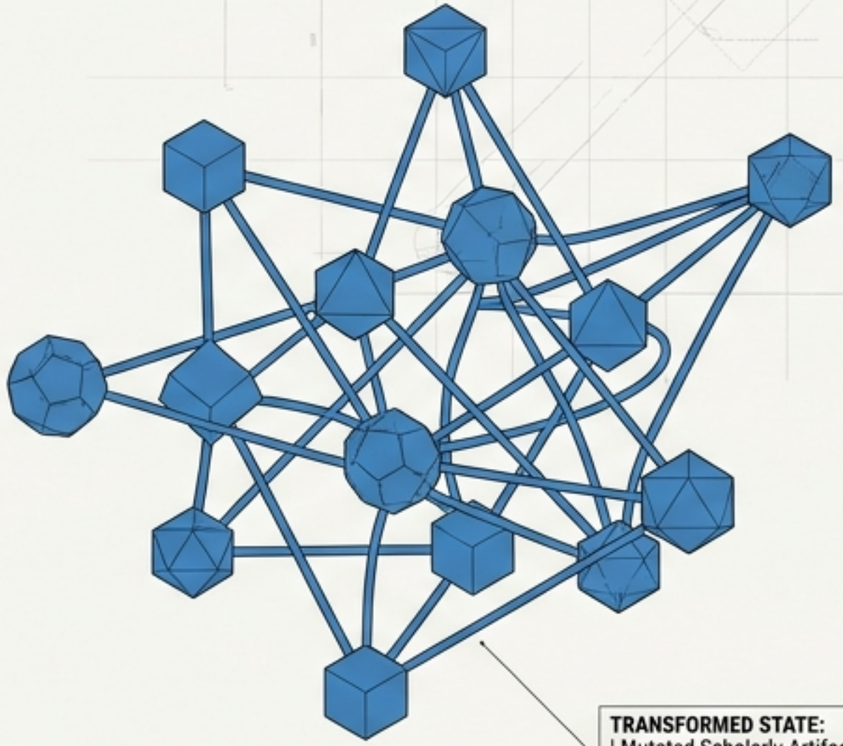
Claimshift: Verification **Under Transformation**

Locators, Claim Graphs, and the Epistemic Boundary of Scholarly Artifacts



INITIAL STATE:
| Formally Verified Matherratical Document [V_i]
| Lattice Structure, Proven Correctness
| Crisp Mint Green = Verified

TRANSFORMATION EVENT | MUTATION BOUNDARY



TRANSFORMED STATE:
| Mutated Scholarly Artifact [M_1]
| Reconfigured Graph, Structural Change
| Architectural Blue = Artifact Under Analysis

SYSTEM DIAGRAM - FIG. 1.0: CLAIMSHIFT TRANSFORMATION PATHWAY

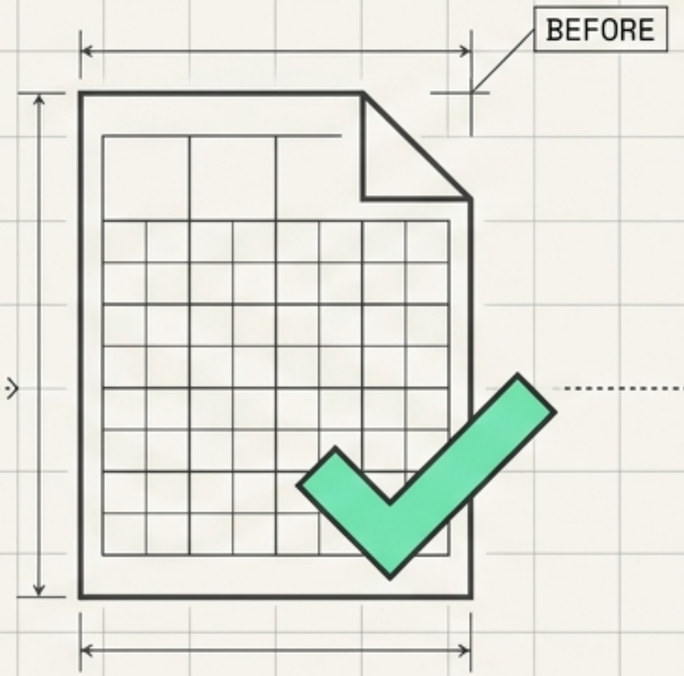
0 10 20 50 100

NOTE: ALL CONNECTIVE INTEGRITY MAINTAINED POST-TRANSFORMATION

PROJECT:		PROJECT: CLAIMSHIFT
		DOCUMENT TYPE: TECHNICAL SLIDE DECK
PROJECT CODE:	DRAWN BY: DATE:	DATE: OCT 26

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Artifacts do not remain fixed. Does verification survive?

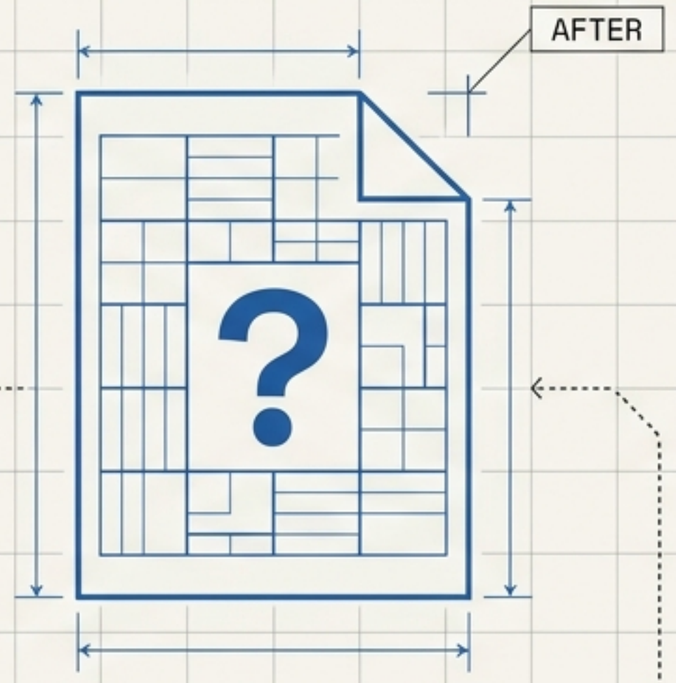


Verified Mathematical Theorem

The Standard Assumption:
Verification is a property of an artifact at the moment it is checked.

TRANSFORMATION_EVENT

(AI summarization, translation, paraphrasing, extraction)



Transformed Artifact

The Reality: Artifacts are constantly reformatted, excerpted, and edited. Truth does not automatically transfer to the successor.

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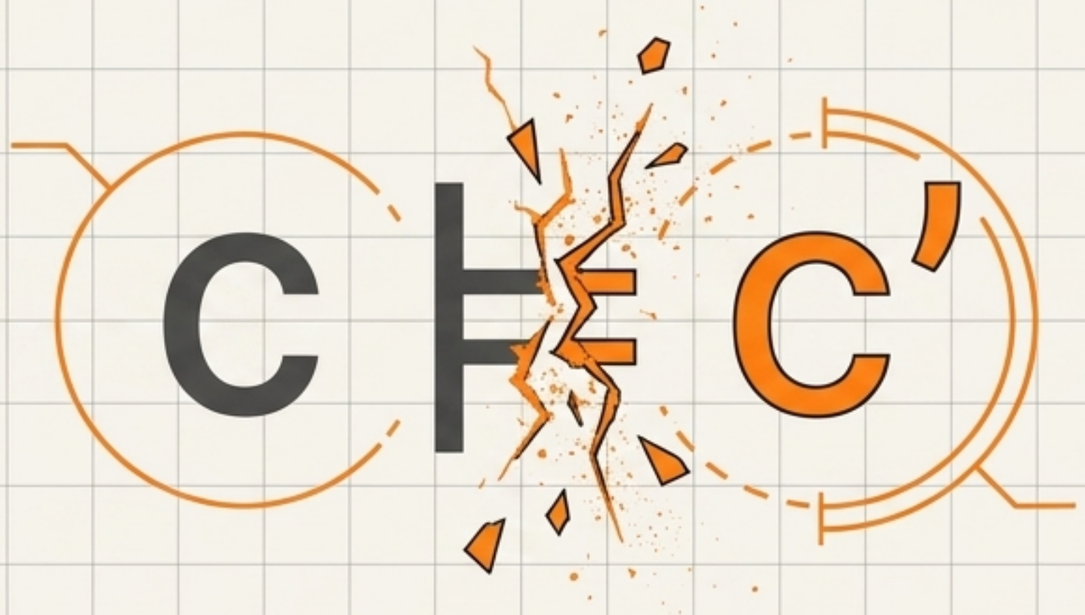
The fatal conflation: Watermarks are not evidence

What we measure



The industry builds provenance ontologies and watermarks. These mechanisms **trace origin, but they cannot inherently prove that the claims made by the earlier artifact are still the claims made by the transformed artifact.**

What we assume

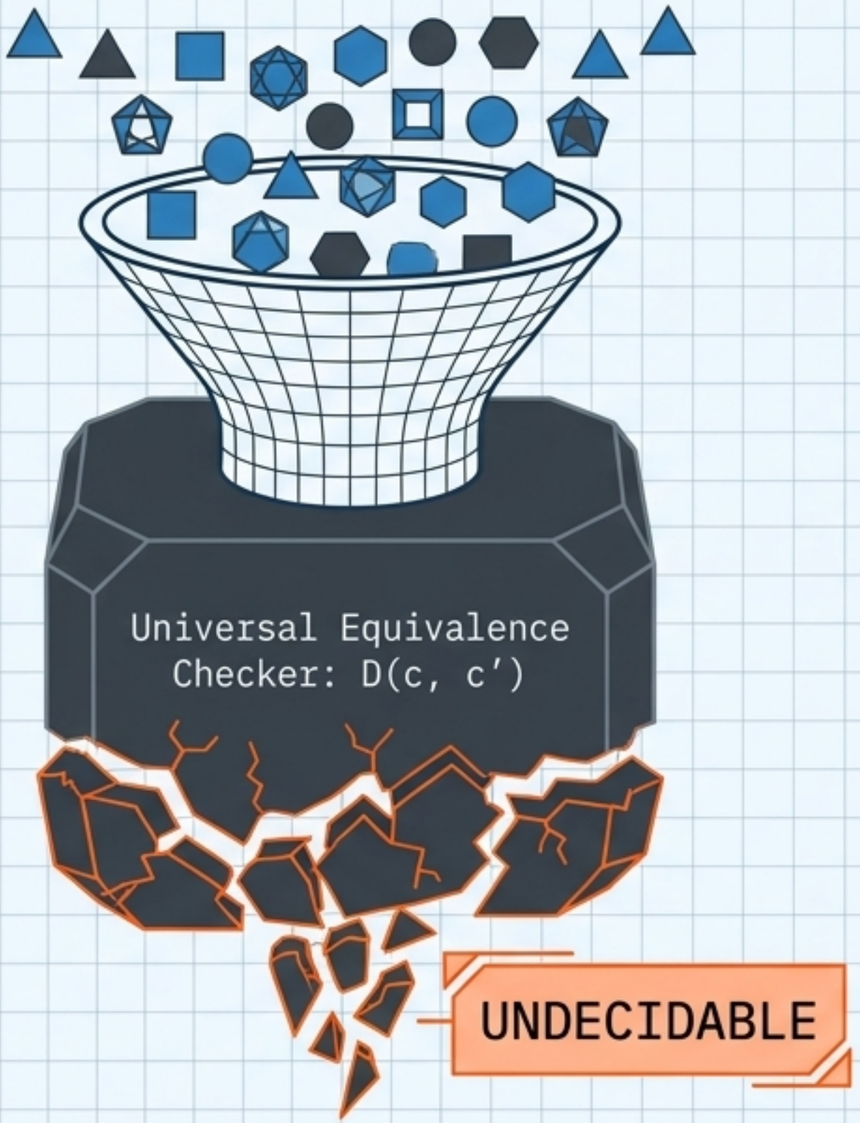


We assume the **semantic meaning** of the original claim remains true. This is a category error.

Persistence of a pointer ≠ validity of the claim.
A robust but semantically stale pointer can authenticate a claim that no longer holds.

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The undecidability of general semantic correspondence



Theorem 1

No total algorithm can decide if $C(c, c')$ holds for all closed Σ-sentences.

The Root Cause

First-order validity over a signature with a relation symbol of arity ≥ 2 is undecidable (Church's Theorem).

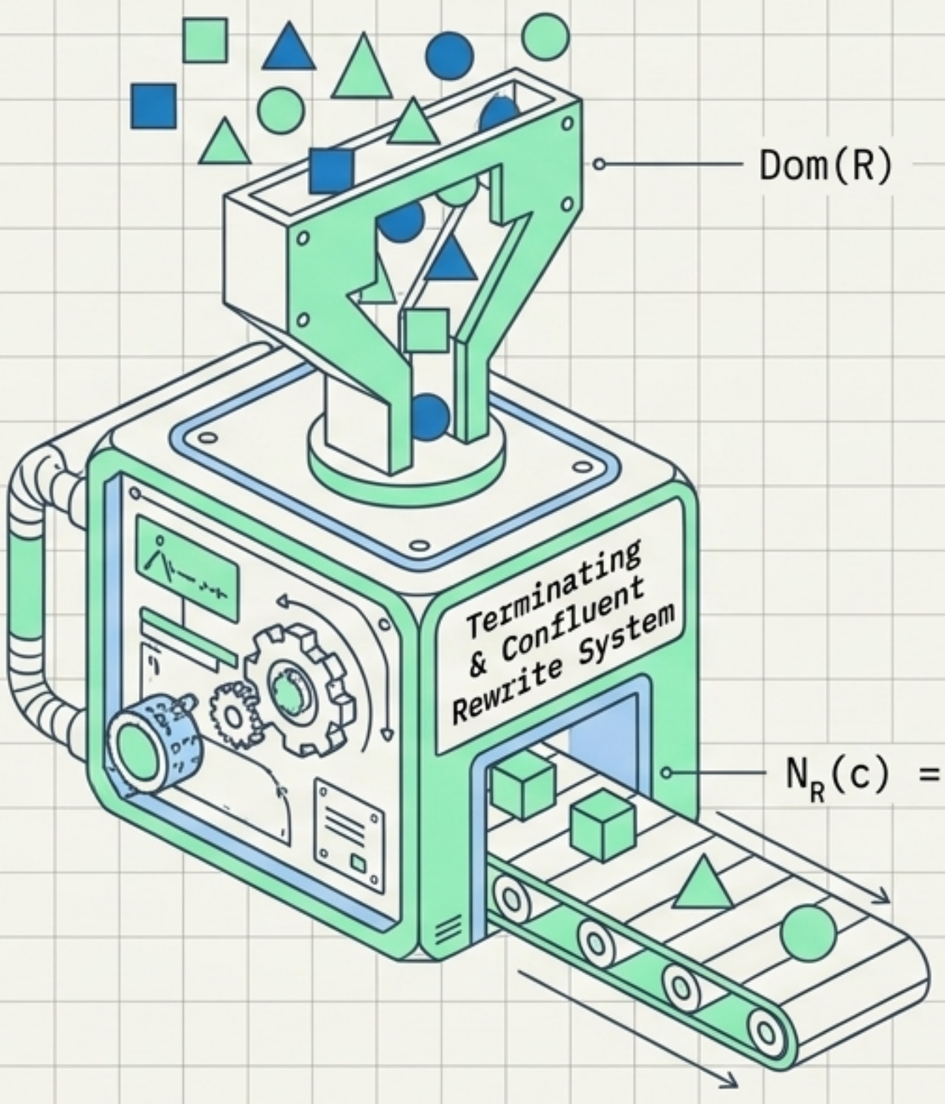
The Implication

You cannot build a universal AI to just "check if the meaning is the same." Any system

claiming universal certification is either incomplete, unsound, or secretly bounded.

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The certified fragment: Correspondence by normalization



Theorem 2

If transformations are restricted to a terminating and locally confluent rewrite system, correspondence becomes decidable by normalization.

Examples of the Fragment

- α-renaming of bound variables (canonical de Bruijn-indexing)
- Commutative reordering of conjuncts
- Double-negation elimination ($\neg\neg\psi \rightarrow \psi$)

The Implication

We don’t need to solve the universe. We just need to mathematically guarantee the benign transformations.

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The three-valued operational boundary

Status Output	Mathematical Condition	What it actually means
PRESERVED	Both claims in fragment, normal forms match.	Mathematically guaranteed to mean the same thing.
NOT_PRESERVED_UNDER_R	Both claims in fragment, normal forms differ.	The specific rewrite rule set failed to establish correspondence. Crucial Note: Does NOT mean false. It just lacks a certificate.
OUTSIDE_CERTIFIED_FRAGMENT	Transformation falls outside the domain.	Not a semantic verdict. An honest admission of inapplicability. Requires human/external check.

> Failure of certification is not certification of failure.

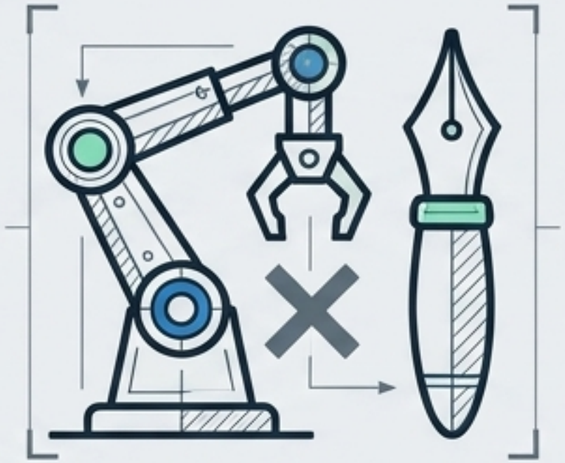
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Four non-negotiable protocol design principles



Locator ≠ Evidence

Robust pointers establish artifact identity, not correctness.



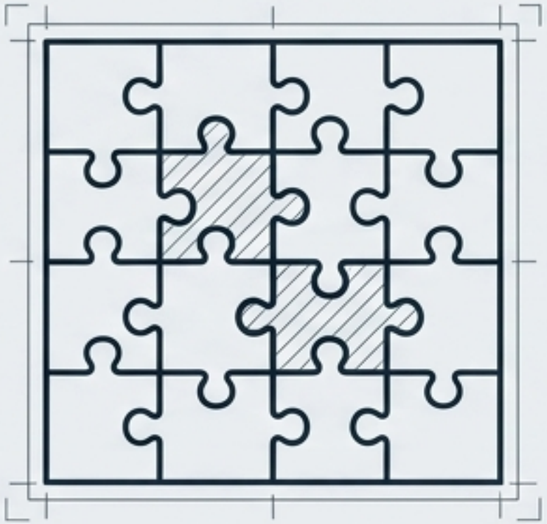
Authorship ≠ Generation

Generative LLM outputs require explicit human assertion signatures.



Verification ≠ Truth

All verification is relative to a specific method, kernel, and environment.

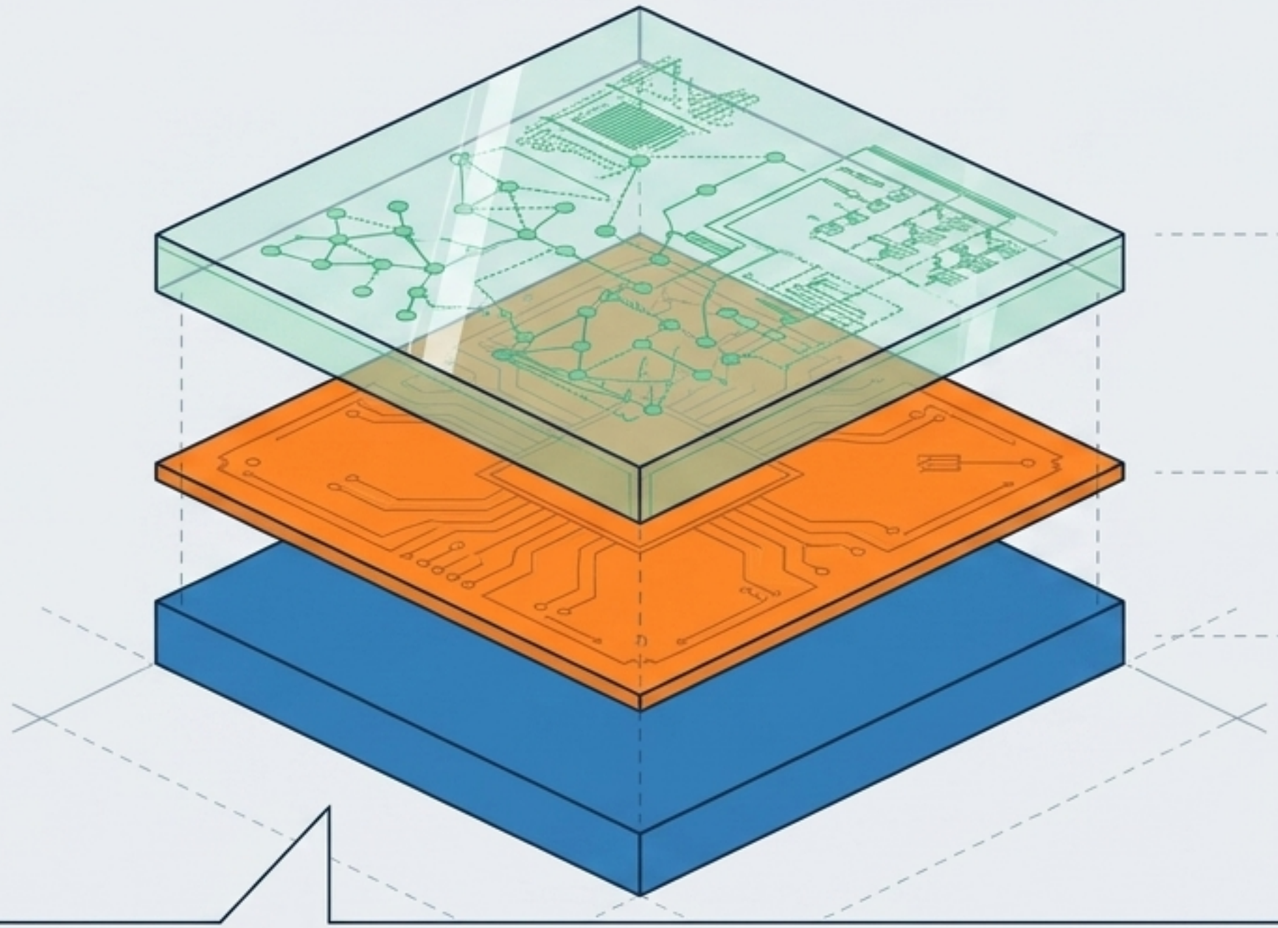


Failure to Certify ≠ Certification of Failure

Never treat OUTSIDE_FRAGMENT as a negative semantic verdict.

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A three-layer separation of concerns



Signed Manifest (M)

An append-only, off-docu containing the full Claim Graph, verification records, and transformation attestations.

Robust Commitment (W(x))

The compact, redundant signal embedded in the artifact (e.g., watermark, C2PA). Contains protocol version, manifest ID, and hash.

The Artifact

The actual PDF, code, or media object the reader encounters.

By separating these layers, a scholarly document can accrue an append-only epistemic history without ever needing to rewrite or republish the original artifact.

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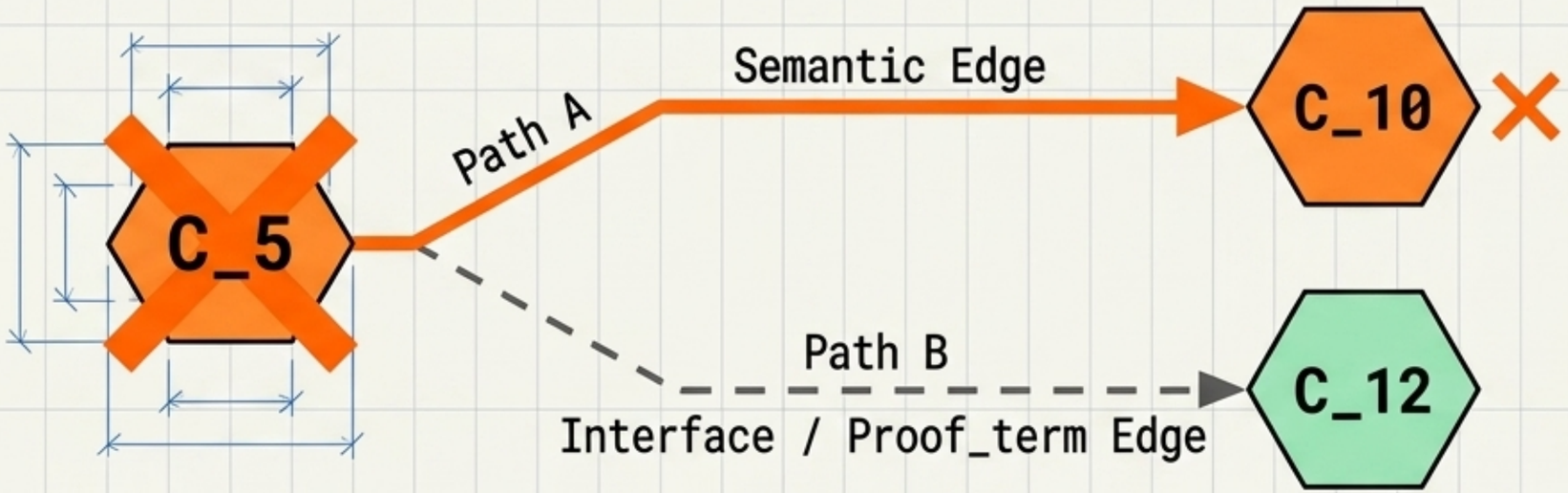
De-conflating signature roles

Role	Mathematical Representation	What it Proves	What it DOES NOT Prove
Assertion	$\sigma_A = \text{Sign}_{-skA}(H(R_i))$	I state this claim is a part of this document.	Verification or generation.
Verification	σ_V over $(H(c), m, I, O, E, r)$	I ran method m on claim c and got result r.	Universal truth, or that the method itself is trusted.
Transformation	$A_{i,T}$ over $(H(c_i), H(c'_i), T, J, r, e)$	I attest c' corresponds to c under adjudication regime J.	Authorization to adjudicate that specific claim type.

Protocol Invariant: No manifest entry may satisfy more than one role without carrying distinct signatures for each.

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Typed dependency propagation: Stopping false revocations



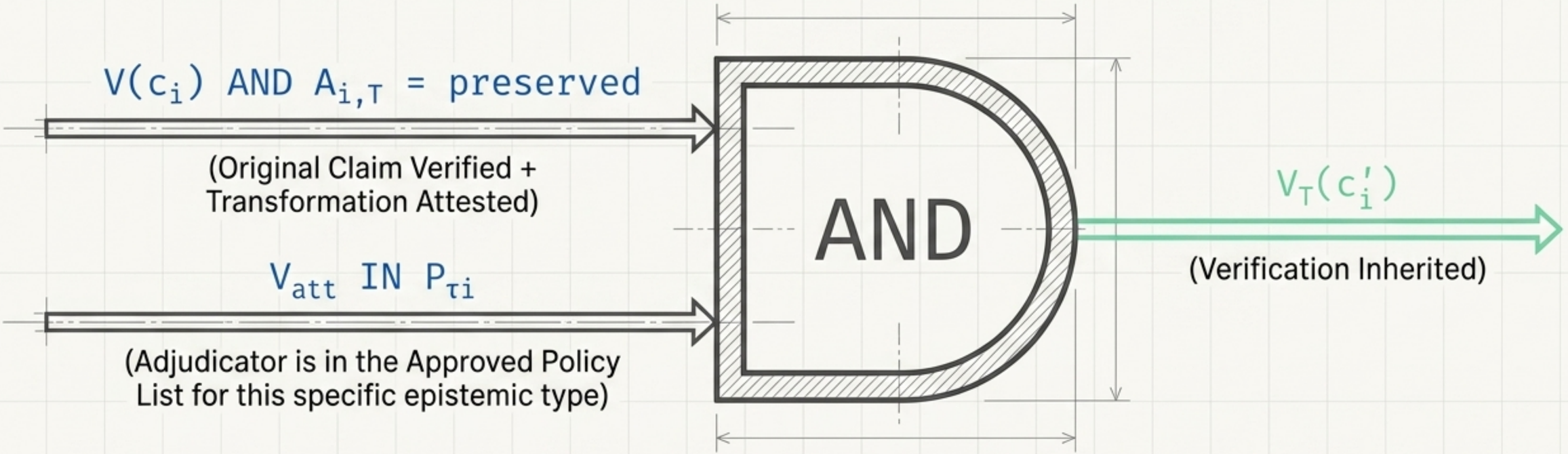
Changing C_5 changes the fundamental proposition C_10 asserts. Downstream failure is required.

C_12 only required C_5's signature or type, not its internal semantic truth. It survives.

Empirical testing revealed that a flat dependency set conflates distinct relations. Claimshift separates semantic dependencies from mechanical ones, ensuring downstream claims are only revoked when their actual truth is threatened.

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The Policy Layer: Trust delegation ≠ Truth



A signature only matters if the policy accepts that adjudicator for that specific claim type (e.g., informal prose accepts expert human attestation, formal math requires a specific proof-assistant kernel).

Different institutions may adopt different policies for the same claim type. Verification inheritance is institution-relative by design.

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The Experimental Setup: A frozen formal specimen

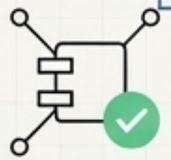
To prove the protocol works, we built a **simplified state transition system (3 locations, 4 primitives, bound-count parity invariant**) that is perfectly verifiable.

This provides a flawless “baseline” artifact against which we can deploy deliberate semantic mutations to observe how the protocol reacts.



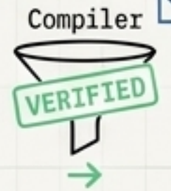
Feature 1: **Zero Dependencies**

Pure Lean 4 core, no Mathlib.



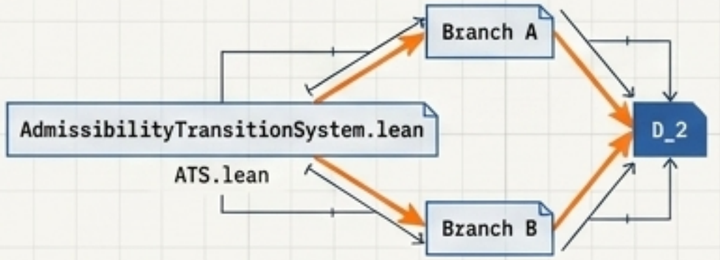
Feature 2: **Zero “sorry” statements**

Kernel-verified clean exit.



Feature 3: **Asymmetric dependency graph**

Two branches sharing a common dependency node D_2.

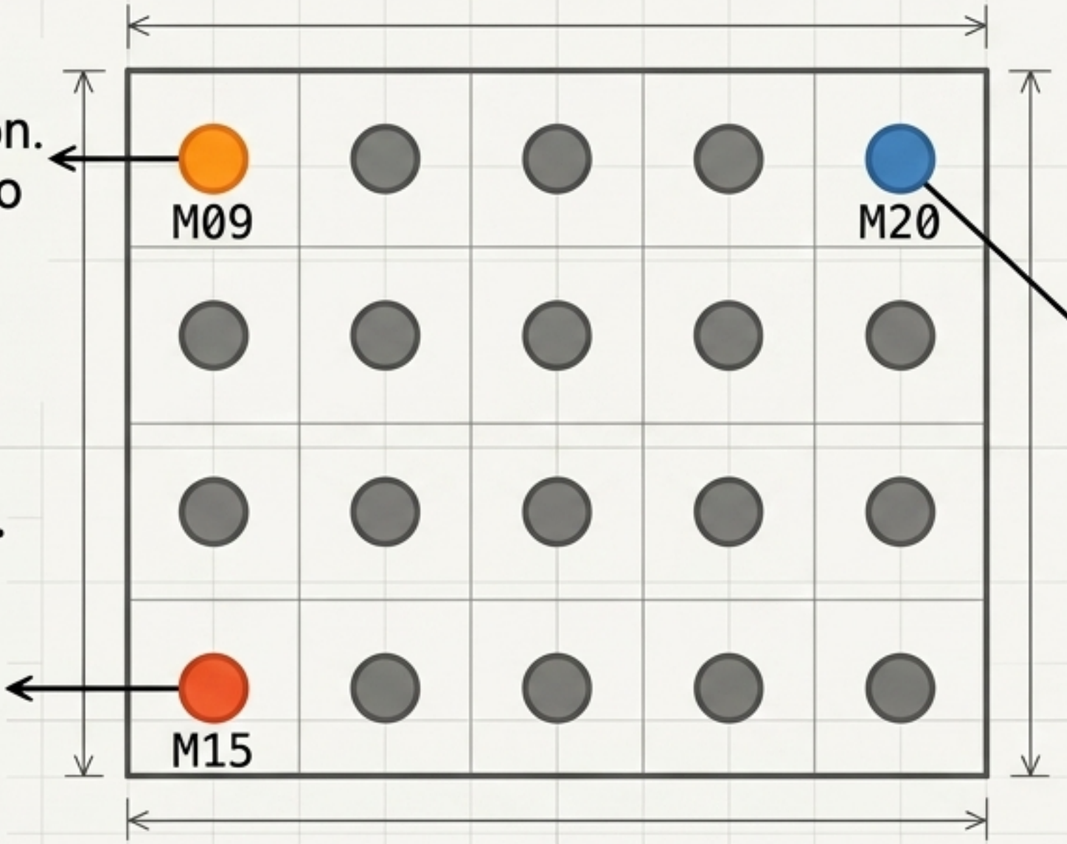


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Ground truth first: The 20-mutant matrix

M09: Constant Mutation.
Isolates branch propagation. Alters a root dependency to ensure only specific branches fail while others survive.

M15: Hypothesis Deletion.
Tests self-refutation. Analyzes if the system catches a mutant that contradicts an existing kernel-checked witness.



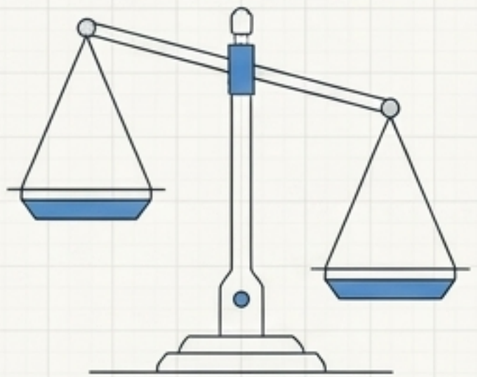
M20: Definition Mutation.
Mid-branch propagation. Tests if a mutation injected halfway down the graph properly cascades without affecting upstream roots.

Preregistered Science: Every ground-truth prediction (G_{-m}) was locked before observation. Ground truth first, then observation—never the reverse.

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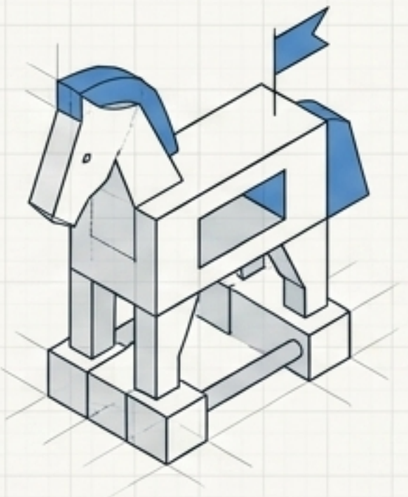
Experimental insights from the Lean 4 audit

Weakening ≠ Easier (M13)



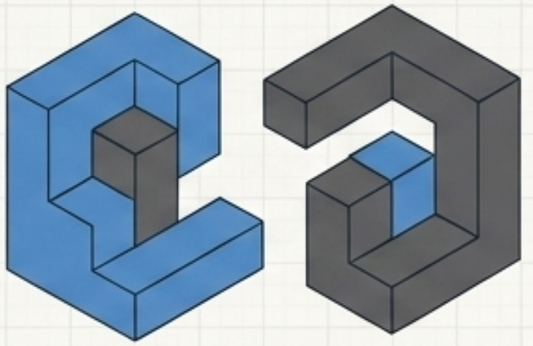
Weakening a hypothesis does not guarantee downstream survival; it can actually make downstream proofs computationally harder or impossible.

Trivial ≠ Safe (M17)



Even injecting tautologically “true” hypotheses can alter formal environments enough to lead to false theorems downstream.

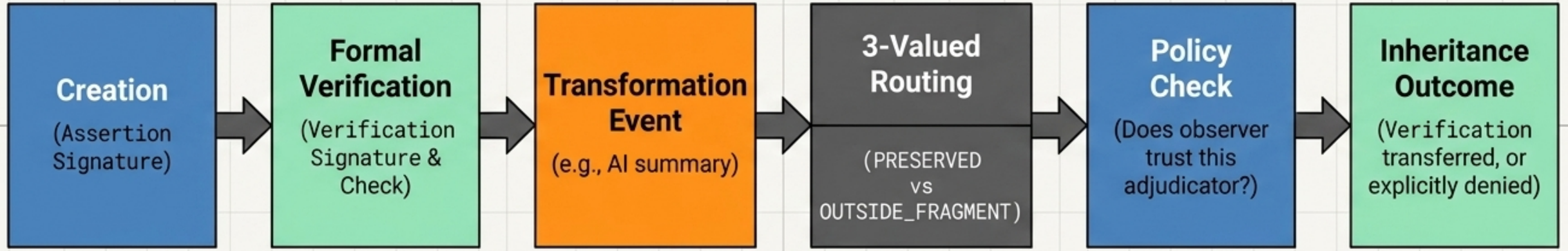
Reprovability ≠ Inheritance (M10)



A transformed claim might still be perfectly provable via different proofs, but that does not mean it inherited the original verification. Logical difference breaks inheritance.

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The Claimshift Framework Overview



Truth does not transfer by default. It must be explicitly re-established or securely inherited.

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